ISE 310: Introduction to Engineering Probability

University of Miami
Fall 2026

Course Details

Class Meeting Time and Location: MWF 10:10AM–11:00AM in Dooly Memorial 217

Instructor: Adam Meyers, Ph.D., axm8336@miami.edu

Office Hours: MEB 284, MWF 11:00AM–12:00PM (after class), or **by appointment**. Feel free to stay after class to ask questions.

Teaching Assistant: Ruimin Yang, rmyang@miami.edu
The TA will only be responsible for grading. You may email the TA with grading questions only, but please CC me to all such emails. You may alternatively email me, and I will speak to the TA as needed.

Prerequisites: MTH 162 or 172 (Calculus II) AND Junior Standing
Primarily, you need to have taken Calculus 1 and 2 and know to work with limits and derivatives and integrals of single and multiple variables.

Textbook: No textbooks are explicitly required for this course. The lecture notes and materials should present what is needed to do homework, exams, etc. However, it is still encouraged that you read or consult textbooks to supplement your understanding. Here are some options for you and ones I have used to help develop the materials for this course:

- **(Primary textbook)** Blitzstein, J. and Hwang, J. (2019). Introduction to Probability (2nd ed.). Taylor & Francis Group, LLC.
    - https://drive.google.com/file/d/1VmkAAGOYCTORq1wxSQqy255qLJjTNvBI/view?usp=sharing
- *(Supplemental)* Ross, S. (2018). A First Course in Probability (10th ed. or earlier). Pearson.
- *(Supplemental)* Walpole, R., Myers, R., Myers, S., and Ye, K. (2017). Probability and Statistics for Engineers and Scientists (9th ed.). Pearson.

Fun Reading: Here are a few books for leisurely reading about probability. These are optional of course.

- Mlodinow, L. (2009). The Drunkard's Walk: How Randomness Rules Our Lives. Vintage Books. ISBN-10: 0307275172.
- Bernstein, P. (1998). Against the Gods: The Remarkable Story of Risk. John Wiley & Sons, Inc. ISBN-10: 0471295639.
- Mosteller, F. (1987). Fifty Challenging Problems in Probability with Solutions. Dover Publications. ISBN-10: 0486653552.

Course Description

This course serves as an introduction to fundamental concepts in probability, particularly for industrial engineering students. Probability is the language for describing uncertainty. For example, if I flip a coin, nobody knows in advance whether heads or tails will occur, but we can model this experiment using the language of probability. Such a model can in turn be used to gain insights about the experiment, such as our expectations of what will happen or a quantification of the variability in results if the experiment were to be repeated. Such insights could then inform later decision making (should we make a bet?). Without probability, the subjects of statistics, data analysis, machine learning, and quantitative analysis in the sciences would be highly limited or impossible. Within industrial engineering, probability is essential for inventory management, financial engineering, service systems, supply chains, logistics networks, simulation, game theory, and decision making under uncertainty. Thus, this course is fundamental for future study within the field, to adequately prepare you for a career in IE and many related fields, and for sharpening your ability to reason when uncertainty is present (which is often the case). Topics covered in this course include the methodological framework of probability, counting, conditional probability, random variables, probability distributions, common discrete distributions, common continuous distributions, mathematical expectation, joint distributions, transformations, moments, and limit theorems.

Course Objectives

- Learn the fundamental concepts of probability useful to engineers, particularly industrial engineers.
- Learn to recognize when uncertainty is present in a given context or application.
- Learn to apply the relevant probability techniques to a given context or application.
- Learn to use a programming language to study probability concepts and/or solve problems.

Grading

Attendance & Participation: 5%

- You will be expected to participate in any in-class activities (e.g., group work) and in any class discussion (e.g., answer questions I pose during class).
- Attendance will be taken every class meeting.
- You will be allowed to miss three class meetings with no penalty. Each subsequent absence will cause a deduction of 0.5% (up to the maximum proportion allotted to attendance/participation) from your attendance/participation grade.
- Please do not email me if you are going to miss a lecture, even if it is an excused absence (e.g., a doctor’s appointment). Part of the rationale behind this attendance policy is to allow you to miss some classes due to valid extenuating circumstances (e.g., doctor’s appointments).
- I also may use your attendance/participation in borderline final grade decisions.

Homework: 25%

- Expect homework to be due roughly every 1-2 weeks. I will notify the class when homework is posted on Blackboard.
- Homework problem statements will be provided in full online (i.e., no textbook is required to view problem statements).
- Homework will be submitted online on Blackboard. Homework will always be due at 11:59 pm on the day it is due. No late homework will be accepted.
- You are permitted and encouraged to work with other students in the class to solve homework problems. You should always attempt problems on your own first before consulting with others. Keep in mind you will need to be fully independent in solving problems to be prepared for exams. (A good metric of being prepared is that you should be able to write solutions from start to finish on a blank sheet of paper with only the problem statement given.) Each student must submit their own solutions “in their own words,” meaning it is not acceptable to submit solutions that are copied in any way from another student. In such a case, any students who engage in cheating will automatically receive an “F” for their final grade. See the section on “Academic Honesty” below for more information.

- Using ChatGPT or other LLMs to provide solutions to homework problems will not help you much where it really counts, i.e., exams. In addition, any suspected cheating from the use of LLMs will be investigated and punished accordingly.
- **You must submit handwritten homework solutions** (either using a physical pencil/pen or a digital pen). It must be neat, legible, and organized; otherwise, we will not be able to grade it and it will receive a zero. There are exceptions to the handwritten policy (e.g., submitting Excel, computer-generated plots, code, etc.). You may also speak with me if you require an exception to this policy.
- Solutions will be provided after the due date has passed.
- Homework assignments may involve completing case studies.
- Homework may require some computer work, e.g., Excel, MATLAB, or Python.

Midterm Exam #1: 20%
- Date: Monday, September 28, 10:00AM – 11:00AM
- Topics: Probability & counting, conditional probability, discrete random variables

Midterm Exam #2: 20%
- Date: Monday, November 2, 10:00AM – 11:00AM
- Topics: Expectation, continuous random variables, moments, joint distributions

Final Exam: 30%
- Date: Wednesday, December 9, 11:00AM – 1:30PM
- Topics: Cumulative (Midterm Exam #1 material + Midterm Exam #2 material + transformations, conditional expectation, inequalities and limit theorems)

<u>Exam Policy</u>

- Exams will be individual, in-class, and closed-book.
- What you should bring to the exam: Pencil and eraser (black pen is permitted but I recommend pencil/eraser), scientific calculator (no graphing calculator or anything more advanced than scientific), and your cane ID card.
- Either I will provide a "cheat sheet" for use during the exam (in which case the cheat sheet will be uploaded to Blackboard in advance of the exam) or I will allow students to make their own cheat sheet with stipulations.
- The exams will be in a "problem-solution" format, meaning you will be given a problem prompt, and you will need to fill out their solutions on blank sheets of paper with only a calculator and formula sheet as aids.
- You are not permitted to use your phone, laptop, tablet, or any other electronic device, except for a basic scientific calculator. In addition, you are not permitted to wear any watches or other devices during the exam; I will display the time on the projector.
- You will not be permitted to use the bathroom during the exam unless the student provides medical documentation or arranges with the instructor in advance of the exam.

<u>Grade Determination</u>

The grading scale used for this course will be <u>no worse</u> than the following:

A+	98%+	B+	87-90%	C+	77-80%	D+	67-70%	F	<60%
A	93-98%	B	83-87%	C	73-77%	D	63-67%		
A-	90-93%	B-	80-83%	C-	70-73%	D-	60-63%		

Course Calendar *(Extremely tentative!!)*

Weeks	**Content**	**Lecture Dates (tentative)**	**Reference chapters/sections**	**Homework Due Dates**
1	Introduction; syllabus	8/19		
1-3	Probability and counting	8/21, 8/24, 8/26, 8/28, 8/31	Blitzstein: Ch. 1	HW #1 – 9/7
3-4	Conditional probability	9/2, 9/4, 9/9, 9/11	Blitzstein: Ch. 2	HW #2 – 9/18
5-6	Discrete Random variables	9/14, 9/16, 9/18, 9/21, 9/23	Blitzstein: Ch. 3	HW #3 – 9/25
6	Flex day (finish lecture material and/or review for Midterm Exam #1)	9/25		
7	Midterm Exam #1	9/28 (10-11AM)	Topics: Probability & counting, conditional probability, discrete random variables	
7-8	Expectation	9/30, 10/2, 10/5	Blitzstein: Ch. 4	HW #4 – 10/12
8-9	Continuous random variables	10/7, 10/9, 10/14, 10/16	Blitzstein: Ch. 5	HW #5 – 10/23
10	Moments	10/19, 10/21	Blitzstein: Ch. 6	HW #6 – 10/30
10-11	Joint distributions	10/23, 10/26, 10/28	Blitzstein: Ch. 7	
11	Flex day (finish lecture material and/or review for Midterm Exam #2)	10/30		
12	Midterm Exam #2	11/2 (10-11AM)	Topics: Expectation, continuous random variables, moments, joint distributions	
12-13	Transformations	11/4, 11/6, 11/9	Blitzstein: Ch. 8	HW #7 – 11/16
13-14	Conditional expectation	11/11, 11/13, 11/16	Blitzstein: Ch. 9	HW #8 – 11/30
14, 16	Inequalities and limit theorems	11/18, 11/20	Blitzstein: Ch. 10	HW #9 – 12/4
16	Flex day (finish lecture material and/or review for Final Exam)	11/30		
16	Final Exam	12/9 (11AM-1:30PM)	Topics: Cumulative (Midterm Exam #1 material + Midterm Exam #2 material + transformations, conditional expectation, inequalities and limit theorems)	

Other Important Dates

I will confirm the exact dates of each midterm exam when we are nearing each exam. The dates listed above are tentative. The final exam date is final, however.

- August 17 – First day of class
- August 26 – Last day for registration and to add a course
- September 7 – No class (Labor Day)
- October 12 – No class (Fall recess)
- November 6 – Last day to withdraw from a course or the semester
- November 23, 25, 27 – No class (Thanksgiving)

- November 30 – Last day of class
- December 16 – Final grades released by faculty in CaneLink by 5PM
- December 18 – Final grades available to students in CaneLink

Classroom Behavior and Inclusiveness

Students are expected to arrive for class on time, stay for the entire allotted time, and to behave in a manner that does not disrupt the learning of others. I expect our class environment to be respectful and inclusive of others. Discussion, questions, and general participation from students is always welcome in this course.

The Use of Electronics in Class

The use of laptops, tablets, and other electronics should be limited to notetaking and using Blackboard (ISE 310) course materials to facilitate learning and participation. If any student uses these devices in ways that distract the students around them or disrupt my ability to communicate effectively in class, that student may be asked to leave the class, or put the device away and take notes by hand instead.

Academic Honesty

All students are bound by the Academic Integrity Policy, which is intended to promote and support an atmosphere conducive to learning. Each student's work is to be their own unless express permission is given by the instructor. Additionally, while students are bound by the policy, faculty are also bound to report violations as offered in the policy.

Members of the university community are expected to be honest and forthright in their academic endeavors. **All work submitted in this course must be your own.** To falsify the results of one's research, to present the words, ideas, data, or work of another as one's own, or to cheat on an examination corrupts the essential process by which knowledge is advanced. If an instructor suspects academic misconduct, such as cheating, plagiarism, or unauthorized collaboration on assignments and/or tests the student will be contacted and if this does not clear up all suspicion the incident will also be reported to the Dean of Undergraduate Studies/Educations. Sanctions for confirmed academic misconduct may include an F in the course. The council may decide on additional actions. If you are unclear about what constitutes academic dishonesty, please see the Academic Integrity Policy:
https://doso.studentaffairs.miami.edu/_assets/pdf/honor-council/the-undergraduate-honor-code.pdf

The Use of ChatGPT and other Generative Artificial Intelligence Software

ChatGPT and other Generative Artificial Intelligence (AI) software may be useful tools for enhancing learning, productivity, and creativity. For instance, they can assist with brainstorming, finding information, and creating materials, such as text, images, and other media. However, these tools must be used appropriately and ethically, and you must understand their limitations. It is important to realize that all AI software has the following limitations:

- How output is arrived at is not clear as the internal processes used to produce a particular output within the generative AI cannot be determined.
- AI output is typically based on data harvested from unknown online sources. As such, it may reflect biases that should be acknowledged. AI output may also be inaccurate or entirely fabricated, even if it appears reliable or factual.

- AI evokes a range of intellectual property concerns; sourcing and ownership of information is often unclear and is currently the subject of ongoing litigation.

If you use AI tools in any part of your work, you are responsible for the final product of that work, both academically and in the workforce.

Principles

- AI should help you think, not think for you. AI tools may be used to help generate ideas, frame problems, and perform research. It can be a starting point for your own thought process, analysis, and discovery. Do not use them to do your work for you, e.g., do not enter an assignment question into ChatGPT and copy and paste the response as your answer.
- The use of AI must be open and documented. The use of any AI in the creation of your work must be declared in your submission and explained. Your faculty can provide guidance as to the format and contents of the disclosure.
- Engage with AI Responsibly and Ethically. Engage with AI technologies responsibly, critically evaluating AI-generated outputs and considering potential biases, limitations, and ethical implications in your analysis and discussions. Ensure that the data used for AI applications are obtained and shared responsibly. Never pass off as your own work generated by AI.
- You are 100% responsible for your final product. You are the user; if the AI tool makes a mistake, and you use it, then it's your mistake. If you don't know whether a statement about any item in the output is true, then it is your responsibility to research it. If you cannot verify it as factual, you should delete it. You hold full responsibility for AI-generated content. Ideas must be attributed, and sources must be verified.
- These principles are in effect unless the instructor gives you specific guidelines for an assignment or exam. It is your responsibility to ensure you are following the correct guidelines. Not following them will result in a breach of the Academic Integrity Policy.
- Data that are confidential or personal should not be entered into generative AI tools. Putting confidential or personal data into these tools exposes you and others to the loss of important information. Therefore, do not do so. See point 3 above.
- The rules and practices on the use of AI may vary from class to class, discipline to discipline. Do not assume that what is acceptable in a Computer Science class will be acceptable in a Philosophy class. It is the student's responsibility to stay informed as to the instructor's expectations. When in doubt, ask.

Most important thing to remember in this class: You should be able to back up anything that you submit as graded work, and heavily relying on tools for at-home assignments (e.g., homework assignments) will not help you sufficiently prepare for the in-class exams (which are done individually without any technology permitted and which account for a large portion of the final grade).

Learning Accommodations and Accessibility

The University of Miami strives to maintain accessibility to all its programs and services. Students with a disability who are experiencing learning difficulties are encouraged to consult the Office of Disability Services (https://camnercenter.miami.edu/disability-services/) to request accommodations. If you think you may have extra challenges with writing assignments, whether because English is not your first language or because your writing skills are weak for other reasons, please utilize the free one-on-one writing assistance at the University of Miami Writing Center (https://writingstudies.as.miami.edu/writing-center/index.html) at the Learning Commons in the Richter Library (phone: 305-284-2956).

Prohibited Discrimination and Harassment Reporting

Any student who has experienced or witnessed sexual assault, relationship violence, sex or gender-based bullying, stalking, and/or sexual harassment may make a report to the Title IX office (contact info below).
University's Title IX Office
Beverly Pruitt, JD, Title IX Coordinator
Maria Sevilla, JD, Deputy Title IX Coordinator
Title IX Office
1320 South Dixie Highway, Suite 100R
Coral Gables, FL 33146
Telephone: 305-284-8624
Email: titleixcoordinator@miami.edu
Website: www.miami.edu/titleix

As instructor of this class, I will direct students who disclose sexual harassment or sexual violence to resources that can help and will only report the information shared to the university administration when the student requests that the information be reported (unless someone is in imminent risk of serious harm or a minor). I am required to report all other forms of prohibited discrimination or harassment to the university administration.

University of Miami employees, including faculty, staff, and TAs, are mandatory reporters of child abuse. This statement is to advise you that your disclosure of information about child abuse to a U Miami employee may trigger the employee's duty to report that information to the university and/or Department of Children and Families.

Additional Resources

Many students at the University of Miami face personal challenges or have psychological needs that may interfere with their academic progress, social development, or emotional wellbeing. The university offers a variety of confidential services to help you through difficult times, including individual and group counseling, crisis intervention, consultations, online chats, and mental health screenings. These services are provided by staff who welcome all students and embrace a philosophy respectful of clients' cultural and religious backgrounds, and sensitive to differences in race, ability, gender identity and sexual orientation.

University of Miami counseling center: https://counseling.studentaffairs.miami.edu/

National Domestic Violence Hotline: Highly-trained advocates are available 24/7/365 to talk confidentially with anyone experiencing domestic violence, seeking resources or information, or questioning unhealthy aspects of their relationship. **Call 1-800-799-7233 or 1-800-787-3224 (TTY).**

988 Suicide and Crisis Lifeline: Formerly known as the National Suicide Prevention Lifeline, the Lifeline provides free and confidential emotional support to individuals in suicidal crisis or emotional distress. **Call or text 988** to be connected to a mental health support specialist. You can also visit www.988lifeline.org to chat with a support specialist online.

Crisis Text Line: **Text HOME to 741741** for free crisis support. Available 24/7. www.crisistextline.org

Local Crisis Services: Banyan Health Mobile Crisis Hotline: **Call 305-774-3616**. Available 24/7.